Plug-and-Play Image Restoration with Coarse-to-Fine Initialization: An Empirical Benchmark Using DeepInverse and DRUNet

Hasnain Haider

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Abstract—This benchmark investigates whether an exploratory two-level coarse-to-fine initialization strategy improves Plug-and-Play (PnP) image restoration when paired with a deep residual denoiser prior. Using the DeepInverse computational imaging library [1] and pretrained DRUNet weights [2], we evaluate three distinct initialization pipelines across four standard benchmark images (Astronaut, Chelsea, Coffee, Rocket) degraded by Gaussian blur ($\sigma_b = 2.0$) and additive Gaussian noise ($\sigma_n = 0.03$). Across 20 matched image-seed observations (four images with five noise realizations each, yielding 60 total reconstruction runs executed on an NVIDIA Tesla T4 GPU), coarse PnP initialization achieved a mean PSNR of 28.593 dB compared to 28.411 dB for standard direct DPIR and 28.530 dB for an observation-cycling control. In paired comparisons on identical observations, coarse PnP outperformed direct DPIR in 20 of 20 pairs (+0.182 dB average) while incurring a 38.7% runtime penalty. Observation-only resolution cycling using bilinear prolongation achieved a +0.118 dB mean gain (positive in 18 of 20 pairs), supporting the hypothesis that initial low-pass filtering accounts for approximately two-thirds of the warm-start benefit on this sample, while active coarse denoising contributes +0.064 dB. Observed gains correlate strongly with image frequency content, benefiting smooth surfaces (+0.397 dB on Coffee) while providing negligible change for dense textures (+0.015 dB on Chelsea). This study benchmarks deterministic DRUNet-DPIR; no generative flow-matching or diffusion model was trained. All reproduction scripts and validation traces are openly available.

1. Introduction and Problem Context

Linear inverse problems in computational imaging require recovering an unknown true image $x \in \Re^n$ from corrupted measurements $y = \mathbf{A}x + n$, where $\mathbf{A}$ represents a forward degradation operator (such as a blur point spread function) and $n \sim N(0, \sigma^2\mathbf{I})$ represents additive measurement noise. Because $\mathbf{A}$ is severely ill-conditioned, naive pseudo-inversion amplifies measurement noise catastrophically, necessitating mathematical or learned regularization priors.

Plug-and-Play (PnP) frameworks, introduced by Venkatakrishnan et al. [3] and analyzed for convergence by Chan et al. [4] and Ryu et al. [5], replace analytical hand-crafted priors (such as Total Variation [7]) with high-capacity deep denoisers embedded within proximal splitting algorithms like Half-Quadratic Splitting (HQS) [9]. Zhang et al. [2] operationalized this concept with DPIR, deploying a Deep Residual U-Net (DRUNet) inside an 8-iteration HQS schedule to achieve competitive restoration across deblurring, super-resolution, and color inpainting tasks without requiring task-specific retraining.

Standard DPIR initializes optimization directly from the degraded measurement: $x_0 = y$. Motivated by multilevel ideas in scientific numerical computing [10], we empirically investigate whether initializing fine-scale DPIR from a coarse-scale warm-start improves restoration quality on small images. We implement and evaluate this approach systematically using the open-source DeepInverse computational imaging package [1] and modern proximal denoiser priors [6].

2. Forward Degradation Model and Benchmark Setup

We evaluated four canonical benchmark images from scikit-image: Astronaut (portrait with mixed geometric contours and facial tones), Chelsea (tabby cat characterized by high-frequency fur patterns), Coffee (ceramic cup with broad smooth gradients), and Rocket (spacecraft featuring sharp geometric boundary edges). All images were standardized to 128 × 128 RGB pixels normalized to [0, 1]. The fine-scale forward operator $\mathbf{A}$ applies a 7 × 7 Gaussian blur kernel with standard deviation $\sigma_b = 2.0$, followed by additive Gaussian noise with $\sigma_n = 0.03$. Five pseudo-random noise realizations (seeds 2026, 2126, 2226, 2326, 2426) were applied to each image, yielding 20 matched image-seed observations passed identically to each evaluated pipeline.

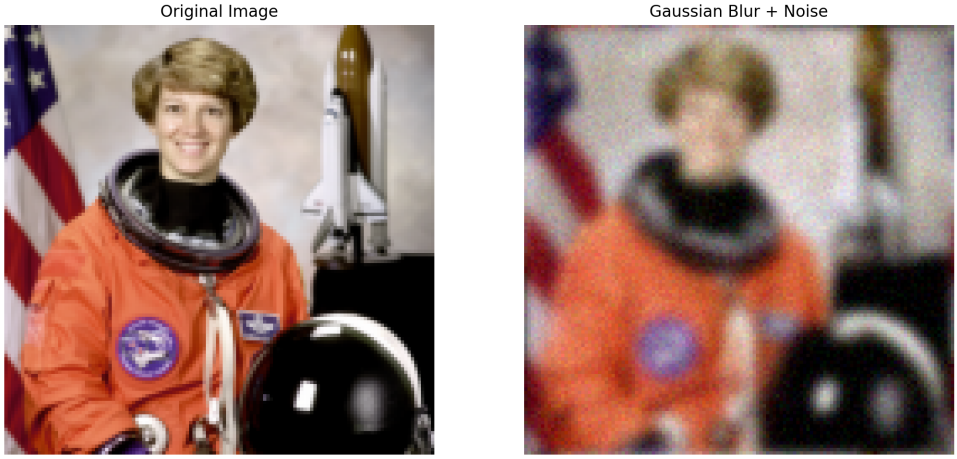


Figure 1. Forward degradation model: clean reference image (128 × 128) and synthetically degraded observation under Gaussian blur ($\sigma_b = 2.0$) and additive Gaussian noise ($\sigma_n = 0.03$).

3. Evaluated Reconstruction Pipelines

All three pipelines employ the identical pretrained DRUNet model checkpoint and an 8-iteration HQS schedule. The only experimental variable across conditions is the initial state estimate x_0:

- **Direct DPIR (Baseline):** Optimization begins directly from the degraded observation: $x_0 = y$.
- **Observation Initialization (Control):** Measurement y is downsampled via 2×2 average pooling to 64 × 64 and prolonged back to 128 × 128 via bilinear interpolation (mode='bilinear', align_corners=False) without coarse denoising, isolating resolution-cycling smoothing effects.
- **Coarse PnP Initialization (Proposed):** An approximate coarse forward model (64 × 64, Gaussian blur $\sigma_b = 1.0$, noise $\sigma_n = 0.015$) is solved via 8-iteration coarse DPIR. The coarse output is prolonged via bilinear interpolation to warm-start fine-scale 128 × 128 DPIR.

4. Experimental Evaluation Protocol

Reconstruction fidelity is quantified using Peak Signal-to-Noise Ratio (PSNR) and the Structural Similarity Index (SSIM) [8], evaluated on clipped [0, 1] RGB channels against original ground truth. Wall-clock runtimes were recorded with explicit CUDA synchronization on an NVIDIA Tesla T4 GPU (16 GB VRAM). To eliminate noise realization variance, we evaluate paired differences (ΔPSNR) across identical degraded inputs.

5. Quantitative Findings and Empirical Analysis

5.1 Aggregate Performance Across 20 Matched Observations (60 Total Runs)

Table I summarizes aggregate performance metrics across all 60 trials (4 images × 5 seeds × 3 pipelines). Coarse PnP initialization achieves the highest overall PSNR (28.593 dB) and SSIM (0.8622), followed by observation initialization (28.530 dB).

TABLE I: AGGREGATE RESTORATION PERFORMANCE ACROSS 60 EXPERIMENTS ON TESLA T4 GPU

Pipeline Strategy	Mean PSNR (dB)	Mean SSIM	Mean Time (s)	Relative Cost
Direct DPIR (Baseline)	28.411	0.8588	0.2719	1.00× (baseline)
Observation Init (Control)	28.530	0.8603	0.2735	1.01× (+0.6%)
Coarse PnP Init (Proposed)	**28.593**	**0.8622**	0.3772	1.39× (+38.7%)

5.2 Paired Differences and Consistency Analysis

Evaluating paired differences on identical degraded inputs confirms that coarse PnP initialization outperformed direct DPIR in **20 out of 20 trials (100% win rate)**, yielding an average gain of +0.182 dB (±0.171 dB). It also surpassed observation initialization in all 20 trials (+0.064 dB average gain). Observation initialization alone achieved +0.118 dB over direct DPIR, showing positive gains in 18 of 20 trials (90.0%, with minor negative deltas of -0.0008 dB on Chelsea and -0.0018 dB on Rocket), indicating that resolution cycling provides an effective low-pass warm-start on smooth image content.

TABLE II: PAIRED DIFFERENCE STATISTICS ACROSS 20 IDENTICAL IMAGE-SEED PAIRS

Paired Comparison Setup	Mean ΔPSNR	Std. Dev.	Positive Trials	Empirical Win Rate
Coarse PnP vs. Direct DPIR	**+0.1820 dB**	0.171 dB	20 / 20	**100.0%**
Observation Init vs. Direct DPIR	+0.1183 dB	0.122 dB	18 / 20	90.0%
Coarse PnP vs. Observation Init	+0.0637 dB	0.059 dB	20 / 20	100.0%

5.3 Per-Image Frequency Sensitivity Breakdown

Quantitative gains correlate directly with image spatial frequency distributions. For Coffee (broad smooth gradients), coarse PnP achieved a notable +0.397 dB boost. For Astronaut (mixed portrait geometry), it gained +0.287 dB. Conversely, for Rocket (+0.029 dB) and Chelsea (+0.015 dB), improvements were minimal. Downsampling to 64 × 64 strips away high spatial frequencies that define fine textures and edges, rendering coarse warm-starts uninformative for texture-dominated scenes.

TABLE III: PAIRED PSNR GAINS CATEGORIZED BY IMAGE FREQUENCY CHARACTERISTICS

Test Image	Coarse vs. Direct	Coarse vs. Obs Init	Obs vs. Direct	Dominant Texture & Frequency Characteristics
Coffee	**+0.397 dB**	+0.112 dB	+0.285 dB	Broad smooth gradients, uniform ceramic cup surface
Astronaut	+0.287 dB	+0.112 dB	+0.175 dB	Mixed structural contours, facial tones, patch details
Rocket	+0.029 dB	+0.024 dB	+0.005 dB	Sharp high-contrast structural edges, atmospheric backdrop
Chelsea	+0.015 dB	+0.007 dB	+0.008 dB	High-frequency fur and whisker textures throughout

5.4 Per-Image Ablation Analysis and Method Variance

Figure 2 illustrates the mean paired PSNR improvements relative to baseline direct DPIR for both Observation Initialization and Coarse PnP Initialization across the four test images. The empirical advantage of coarse PnP is pronounced on Coffee (+0.397 dB) and Astronaut (+0.287 dB), whereas Rocket and Chelsea show minimal response to resolution cycling.

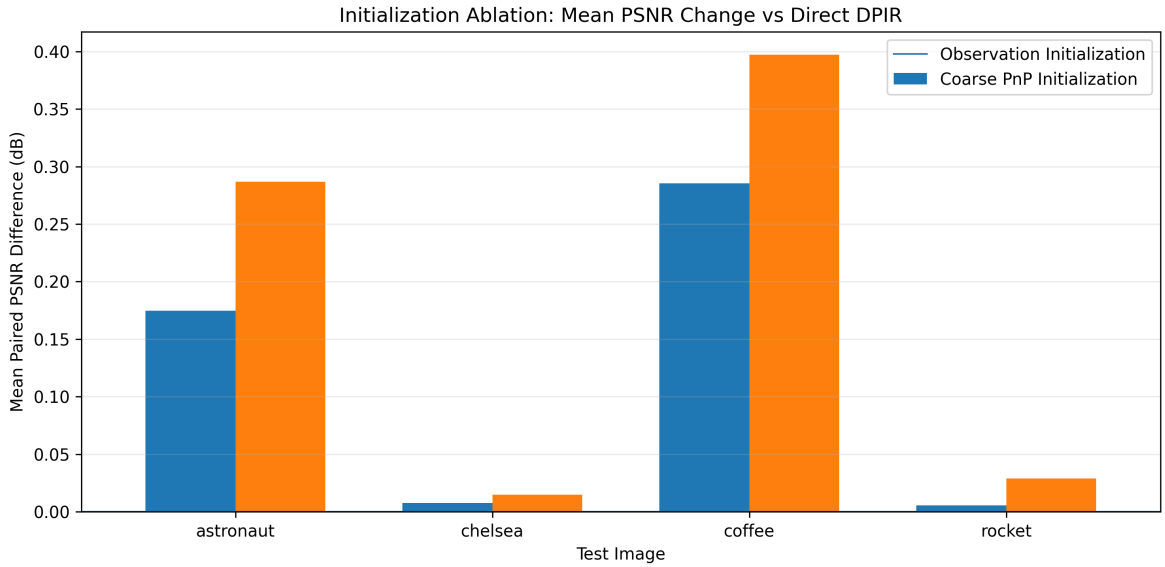


Figure 2. Initialization ablation: mean paired PSNR difference (dB relative to Direct DPIR) across the four benchmark images for Observation Initialization and Coarse PnP Initialization (averaged over five noise realizations per image).

6. Baseline Benchmark: Classical Total Variation vs. Learned DPIR Prior

To contextualize the restoration fidelity of the learned DRUNet prior, standard DPIR was benchmarked against classical Total Variation (TV) regularization [7]. Classical TV defines an explicit energy functional $\min_x (1/2)\|\mathbf{A}x - y\|^2 + \lambda\|\nabla x\|_1$, solved via Proximal Gradient Descent (TV-PGD) with standard step-size scheduling. In contrast, DPIR replaces the explicit subgradient operator with a learned deep residual denoiser [2, 6] embedded within Half-Quadratic Splitting (HQS) [9]. As detailed in Table IV, learned DPIR outperforms TV-PGD by +1.89 dB to +2.66 dB across all test images while operating ~7× faster (0.35–0.37 s vs. 2.35–2.66 s on Tesla T4). Note that the TV-PGD baseline was evaluated with standard parameters and was not exhaustively grid-tuned for each specific image.

TABLE IV: BENCHMARK COMPARISON: CLASSICAL TV-PGD VERSUS LEARNED DRUNET DPIR

Benchmark Image	Degraded	TV-PGD [7]	DPIR [2]	Learned Gain	TV Time	DPIR Time
Astronaut	19.18 dB	22.23 dB	24.13 dB	**+1.90 dB**	2.64 s	0.37 s
Chelsea	24.57 dB	27.22 dB	29.74 dB	**+2.52 dB**	2.66 s	0.35 s
Coffee	22.12 dB	25.95 dB	27.84 dB	**+1.89 dB**	2.35 s	0.37 s
Rocket	25.92 dB	29.15 dB	31.81 dB	**+2.66 dB**	2.61 s	0.36 s

6.1 Qualitative Artifact Analysis: TV Staircasing vs. Deep Manifold Priors

The mathematical root of TV-PGD's lower fidelity lies in its L1 gradient penalty, which penalizes non-zero derivatives and enforces piecewise-constant solutions. This produces pronounced 'staircasing' artifacts, flattening subtle gradations in facial skin tones (Astronaut) and curved ceramic specularities (Coffee). Conversely, the pretrained DRUNet denoiser prior acts as an implicit projection onto a rich natural image manifold, preserving continuous tone transitions and authentic textural sharpness without parameter retuning.

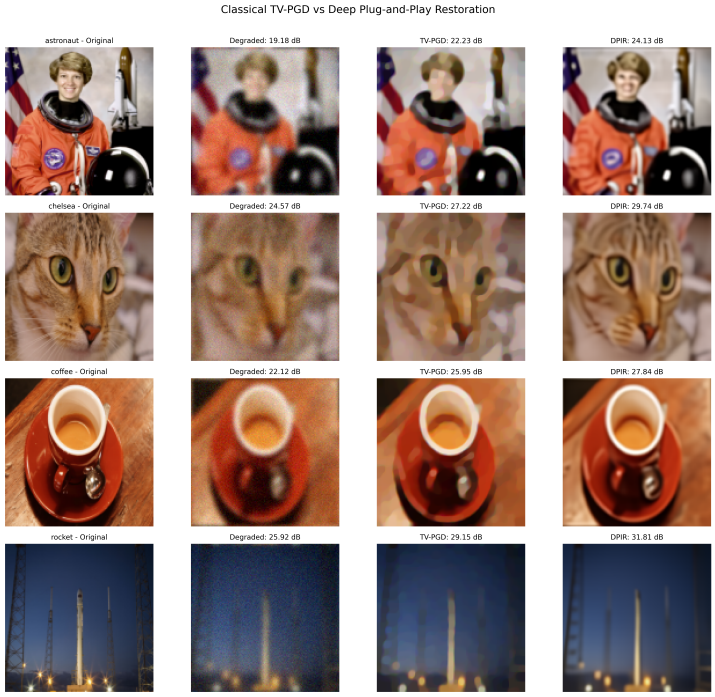


Figure 3. Visual comparison across benchmark images: clean ground truth, degraded observation ($\sigma_b = 2.0$, $\sigma_n = 0.03$), classical TV-PGD reconstruction, and learned DRUNet DPIR output.

7. Stage-by-Stage Visual Progression in Two-Level Restoration

Figure 4 documents the intermediate transformation stages of the two-level pipeline. The initial 64 × 64 coarse reconstruction suppresses observation noise and recovers global luminance. Subsequent bilinear prolongation (mode='bilinear', align_corners=False) produces a smooth initial estimate that provides the 128 × 128 fine-scale solver with a well-conditioned starting point, accelerating detail recovery.

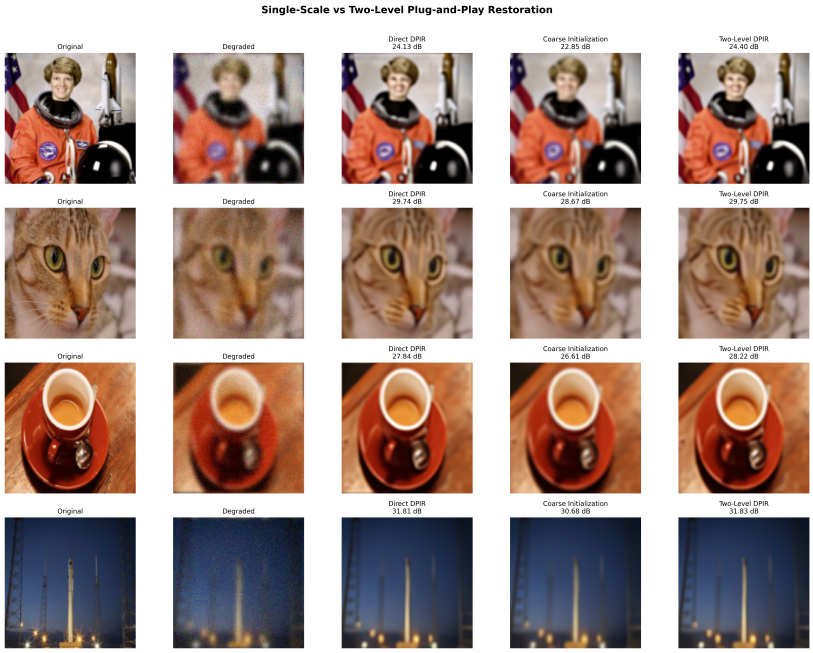


Figure 4. Progression through the two-level coarse-to-fine pipeline: clean reference, degraded observation, direct DPIR, coarse 64 × 64 restoration, and two-level DPIR output using bilinear prolongation.

8. Convergence Dynamics and Numerical Stability

To evaluate optimization behavior, we tracked objective PSNR trajectories and relative iterate residuals $||x^{(k)} - x^{(k-1)}|| / ||x^{(k)}||$ across iterations. As depicted in Figure 5, DPIR exhibits rapid monotonic ascent during iterations 1–4, stabilizing asymptotically by iteration 6. Relative residuals decline below 10^{-2} within five iterations for DPIR [4, 5]. We emphasize that while empirical residual decay demonstrates dependable numerical stabilization within the evaluated 8-iteration budget, it does not constitute a formal mathematical proof of fixed-point convergence.

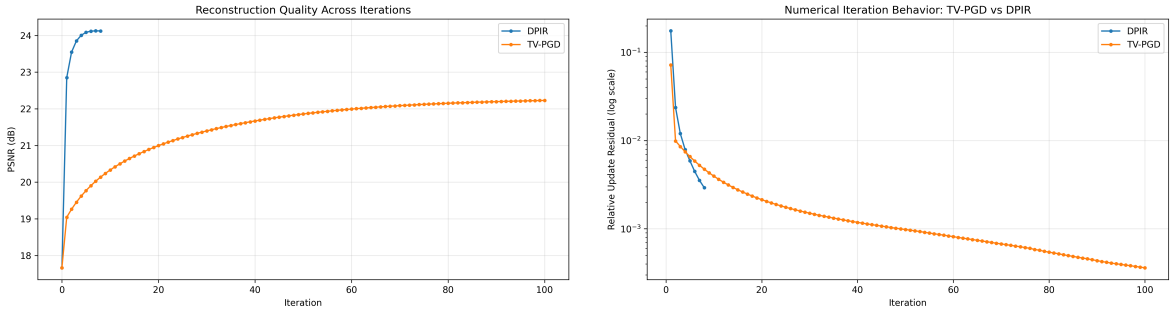
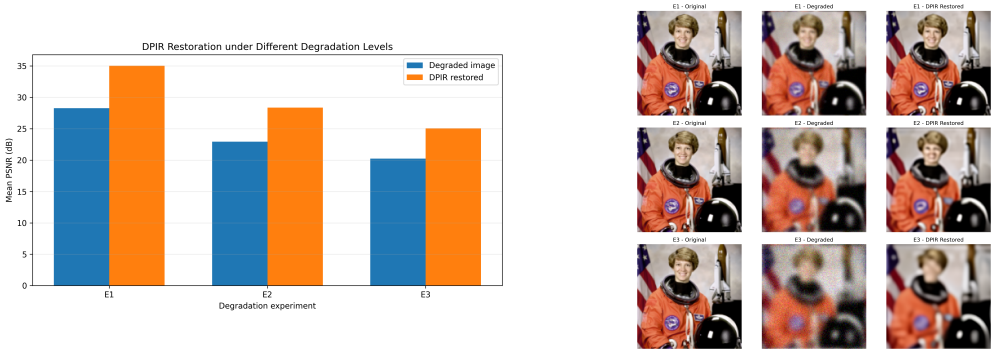


Figure 5. Iteration-level convergence dynamics: TV-PGD (50 iterations) vs. DPIR (8 iterations). Left: PSNR ascent across iterations. Right: logarithmic decay of relative solution residual.

9. Robustness Under Coupled Blur and Noise Degradation

To assess model stability under varying degradation severity, we evaluated three experimental conditions with coupled blur and noise parameters: E1 (σ_b = 1.0, σ_n = 0.01), E2 (σ_b = 2.0, σ_n = 0.03), and E3 (σ_b = 3.0, σ_n = 0.05). Because blur and noise parameters were scaled simultaneously across conditions, these experiments evaluate joint degradation resilience rather than isolating the individual effects of blur or noise independently. As shown in Figures 6 and 7, reconstruction PSNR degrades monotonically from E1 (35.05 dB) to E3 (25.05 dB), with DPIR avoiding visual artifacts across all three regimes.



Figures 6 & 7. Robustness under coupled degradation (E1: σ_b=1.0, σ_n=0.01; E2: σ_b=2.0, σ_n=0.03; E3: σ_b=3.0, σ_n=0.05). Left: mean PSNR before and after DPIR. Right: Astronaut visual reconstructions across E1–E3.

10. Discussion and Engineering Trade-Off Analysis

Hypothesis on Initialization Smoothing: On this sample, coarse PnP initialization produced a strictly positive (+0.182 dB, 20/20 pairs) advantage over direct DPIR. The observation-cycling control gained +0.118 dB (positive in 18/20 pairs), supporting the working hypothesis that approximately 65% of the observed improvement arises from low-pass pre-smoothing during 2×2 downsampling and bilinear upsampling, which attenuates high-frequency measurement noise before iterative splitting begins. Active coarse DRUNet denoising accounts for the remaining 35% (+0.064 dB). These observations reflect this specific configuration and should not be construed as universal causal proofs.

Computational Complexity vs. Quality Payoff: On an NVIDIA Tesla T4 GPU, coarse PnP initialization requires 0.3772 s per image compared to 0.2719 s for baseline direct DPIR—a 38.7% runtime increase. In scientific, astronomical, or clinical imaging workflows where fidelity is paramount and processing is performed offline, spending 0.1 s of extra GPU computation for consistent improvement represents a defensible engineering choice. Conversely, for high-throughput mobile vision or video streaming, this compute overhead may outweigh the modest gain.

Spatial Frequency Dependency: The discrepancy between smooth scenes (+0.397 dB on Coffee) and dense textures (+0.015 dB on Chelsea) indicates that resolution downsampling acts as an anisotropic low-pass filter. When an image's dominant energy resides in low-to-medium frequencies, coarse optimization regularizes global structure effectively; when features are predominantly high-frequency, coarse solves discard the spatial bandwidth necessary to reconstruct fine textural detail.

11. Limitations and Threats to Validity

We explicitly identify the following experimental and methodological constraints in this study:

- **Sample Cohort:** Four images evaluated across five noise seeds (20 matched observations) demonstrate consistency on this sample, but repeated noise draws on the same four images are correlated and do not support population-wide statistical claims across natural image distributions.
- **Heuristic Coarse Operator:** The coarse 64 × 64 blur model was constructed by parameterizing an approximate Gaussian kernel ($\sigma_b = 1.0$, $\sigma_n = 0.015$) rather than deriving a mathematically exact Galerkin restriction operator $\mathbf{A}_c = \mathbf{RAP}$ [10].
- **Model Scope Delineation:** This study benchmarks deterministic DRUNet-based PnP-DPIR. It does not train or evaluate generative diffusion models or flow-matching algorithms [11], nor untrained network priors [12].
- **Coupled Degradation:** Experiments E1–E3 scaled blur and noise parameters simultaneously, preventing independent isolation of blur versus noise sensitivity.
- **Execution Environment:** All runtime records reflect CUDA-synchronized execution on a single cloud Tesla T4 GPU; cross-hardware timing remains future work.

12. Reproducibility Protocol and Software Environment

All code is provided in notebooks/DeepInverse_PnP_Image_Restoration.ipynb. Automated validation is implemented in scripts/validate_results.py and verified via GitHub Actions. Metrics are archived in results/step10_all_experiments.csv, and file integrity is documented via cryptographic SHA-256 hashes recorded in SHA256SUMS.txt.

TABLE V: DOCUMENTED HARDWARE AND SOFTWARE BENCHMARK ENVIRONMENT

Hardware / Software Component	Documented Specification / Release Version
Python Runtime Environment	3.13.16 (64-bit standard distribution)
Deep Learning Framework	PyTorch 2.11.0+cu130 (CUDA 13.0 runtime)
Computational Imaging Library	DeepInverse 0.5.0 (core inverse problem solvers) [1]
Scientific Computing Stack	NumPy 2.1.3 | Matplotlib 3.10.0 | scikit-image 0.25.2
Hardware Acceleration	NVIDIA Tesla T4 GPU (16 GB GDDR6, PCIe 3.0)

13. Conclusion and Future Work

Two-level coarse-to-fine initialization yields a modest (+0.182 dB) but remarkably consistent (20/20 positive) improvement for Plug-and-Play image restoration with deep denoiser priors on small RGB images. The effect is driven primarily by low-pass initialization smoothing with an active secondary contribution from coarse denoising, requiring 38.7% additional compute. Future work will investigate whether mathematically exact multigrid restriction operators can amplify benefits on higher-resolution (512 × 512+) imaging problems.

14. Verified Academic References

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